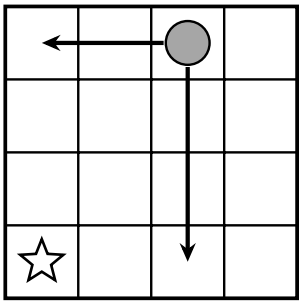
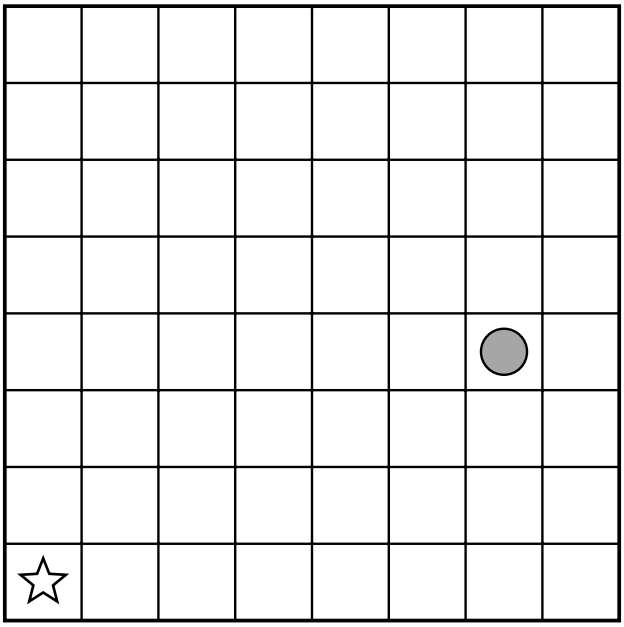


Players take turns. On a board, a move slides the token straight left or straight down, toward the star's side, one or more squares. Whoever moves the token onto the star wins.

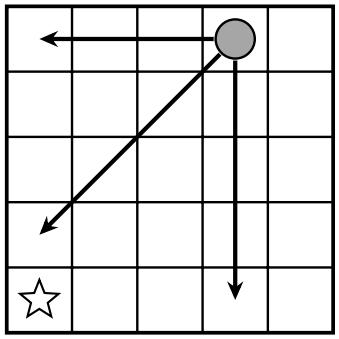
In a game with piles of counters, a move takes one or more counters from one pile. Whoever takes the last counter wins.

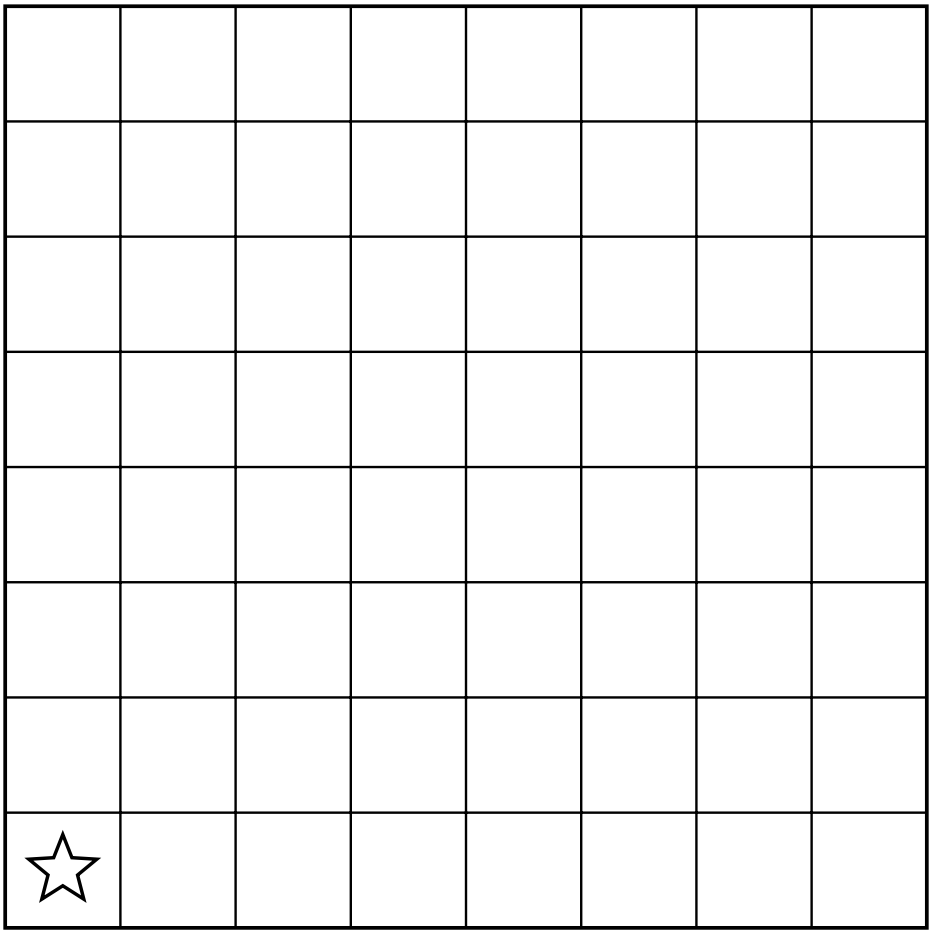
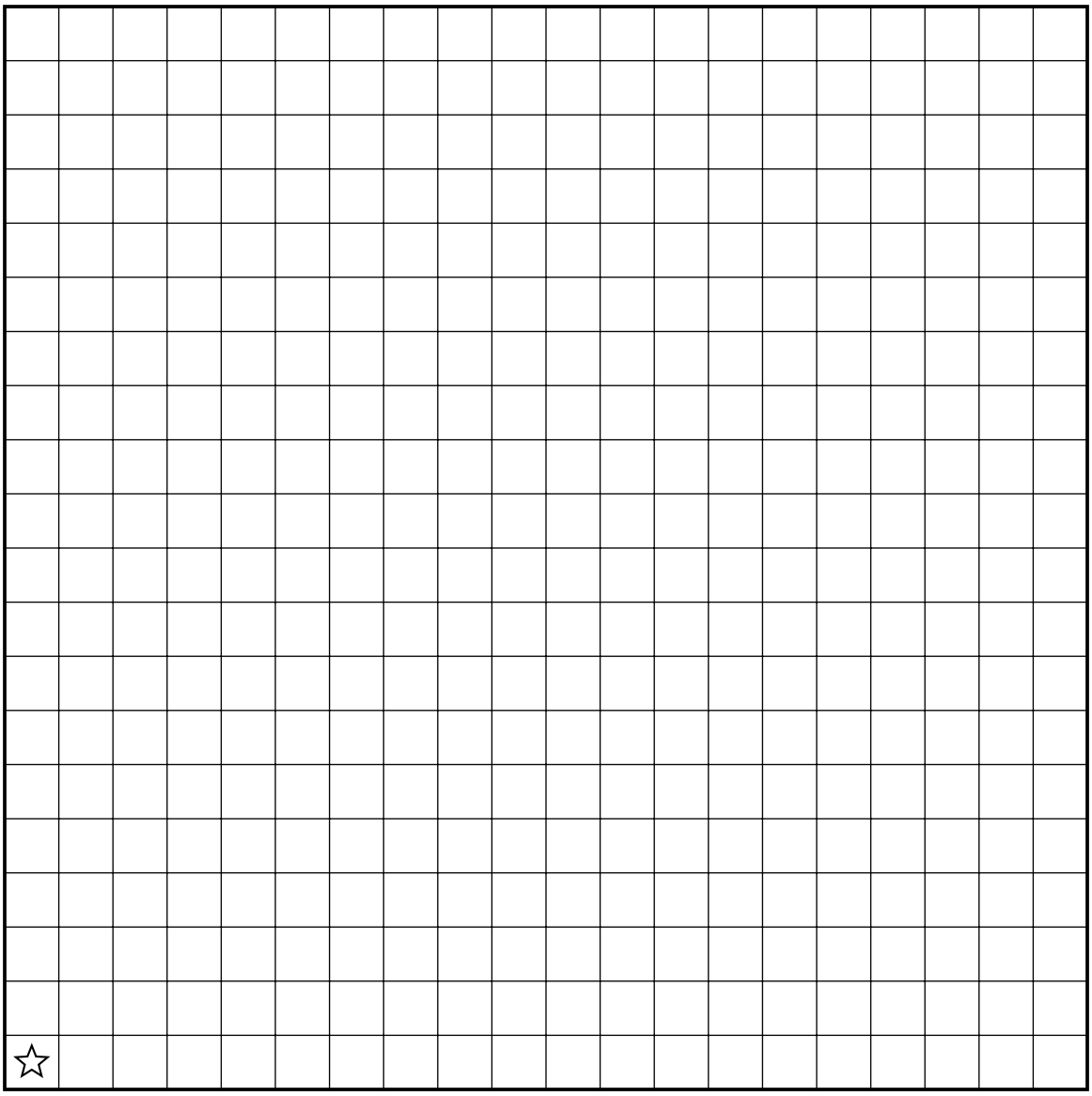
Problem 2: Find two piles of counters that make the same game as this board with the token on the dot. Explain how each move on the board matches a move with the piles. Which player can always win from piles of 23 and 17? From piles of 30 and 30?



Problem 7: Explain why your rule from Problem 5 is right for every start, even with piles of hundreds of counters.

Problem 8: In a new game, the token may also slide diagonally down and to the left, one or more squares. Color every square of this 8-by-8 board except the star: green if the first player can always win when the token starts there, red if the second player can always win.





Problem 9: In the game from Problem 8, mark all the red squares on this 20-by-20 board. Explain how you know you have found them all.